DHVANI Pod – What to Measure and How

Oct 4, 2026 · @harshit

Already fixed

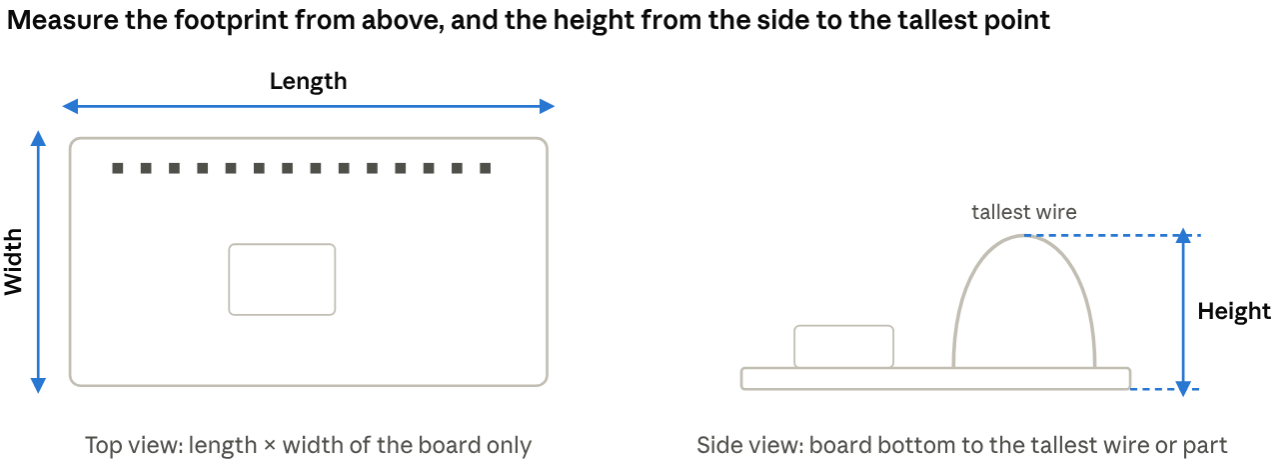
These are settled — no need to measure again.

Item	Value
Nucleo footprint	140 × 76 mm (as measured, overhang included)
Nucleo height	Bottom of the board to the tallest jumper wire — fits inside 100 mm
Inside height of the pod	100 mm
Clearance the design adds	+1.5 mm on each side of every pocket, +10 mm between parts and walls
Printer	Bambu Lab H2D — the pod must stay under about 300 mm in its longest direction

Measure the **real** size of each part. Don't add margins yourself — the design adds them.

Tools and rules

A ruler is enough; calipers are better for the small holes. Give every number in millimetres.



how to measure any board · footprint from above, height from the side

- **Footprint:** measure the board itself from above, edge to edge.

- **Height:** from the very bottom of the part to its highest point — wires, chips, capacitors or connectors, whichever sticks up most.
- **Round to 1 mm** for boards and the battery; **0.5 mm** for anything that goes through a hole (pot bushing, cables, USB plug).
- **Measure the real thing,** not the datasheet — your breadboard with its circuit is what has to fit.
- **Take a photo of each part with the ruler beside it.** It catches mistakes in the numbers.

Must measure: breadboard and battery

These two set the size of the whole pod, so measure them first.

1. Breadboard (with the TL074 circuit built on it)

Measure	How
Length	Long edge of the breadboard itself
Width	Short edge
Height	Bottom of the breadboard to the tallest thing on it — TL074, capacitors, or the highest jumper wire
Wire overhang	Do any wires stick out past the edges? If yes, how far from the edge (mm)

A half-size breadboard is usually about 83 × 55 mm, a full-size one about 165 × 55 mm — check which you have. A full-size board changes the layout, because 140 + 165 mm side by side won't fit the printer bed.

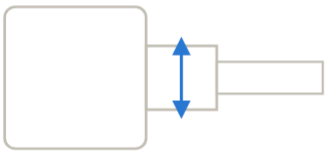
2. 9 V battery (with the clip on)

Measure	How
Length	From the battery's bottom to the top of the clip
Width × height	The battery's two short sides (usually about 26 × 17 mm)
Clip wire length	How long the clip's wires are — so the battery sits close enough to the TL074

Hole sizes: dial, probe cables, USB plug

These set the hole sizes. Measure to 0.5 mm — a hole that is 1 mm too small means reprinting.

For anything that passes through a hole, measure its widest part



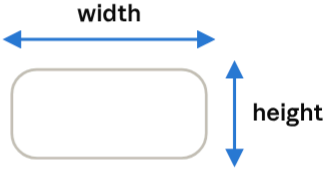
Threaded bushing diameter

Potentiometer, side view



connector width x height

Probe cable with its connector



USB-C plug, end view

hole sizes · the widest part has to pass

3. Potentiometer (clarity dial)

Measure	How
Bushing diameter	The threaded collar the nut screws onto (usually 7 mm) — this is the lid hole
Bushing length	How far the thread sticks out — the lid must be thinner than this so the nut can grip
Knob diameter	Only if you fit a knob — so it clears the lid window

4 and 5. TDS probe cable and DS18B20 cable

Measure	How
Cable thickness	Thickness of the cable itself
Connector width x height	The plug at the end, if it must pass through the hole. If you can unplug it inside the pod, write: unplugs

6. USB-C cable plug

Measure	How
Width x height	The plastic shell of the plug, not the metal tip — the current slot is 30 x 12 mm

Optional: small parts

Only needed if a part will sit loose in the pod rather than on the breadboard or hanging on its wires — then it gets its own small pocket.

#	Part	Measure
7	INA219 board	Length × width × height, including the screw terminal
8	TDS module board	Length × width × height
9	Buzzer	Diameter × height — and whether you want a sound hole in the wall next to it

If a part sits on the breadboard, just write "on breadboard" — its height is already in the breadboard's height.

Decisions to make

Pick one option for each — the design follows from these.

Decision	Options	Current design
Shape	Rounded box · cylinder (looks like an AUV hull section)	Rounded box
Layout	Nucleo and breadboard side by side · breadboard stacked on a shelf above the Nucleo	Side by side
Dial position	On the lid · on a side wall	Lid
Lid fixing	4 M3 screws · snap fit	M3 screws
Material	PLA (easiest) · PETG (tougher, better near water)	PLA
Colour	Your filament colour, for the video	—

If the breadboard is full-size, side by side won't fit the bed — choose the stacked layout or a two-part pod.

Fill in and send back

Copy this table, fill the blanks, and send it with one photo of each part next to a ruler.

#	Part	Measurement	Your value (mm)
—	Nucleo	Footprint / height	140 × 76 / fits in 100 (done)
1	Breadboard	Length × width × height	
1	Breadboard	Wire overhang past the edges	
2	9 V battery + clip	Length × width × height	
3	Potentiometer	Bushing diameter / bushing length / knob diameter	
4	TDS cable	Cable thickness / connector size or "unplugs"	
5	DS18B20 cable	Cable thickness / connector size or "unplugs"	
6	USB-C plug	Width × height of the plastic shell	
7	INA219	L × W × H, or "on breadboard"	
8	TDS module	L × W × H, or "on breadboard"	
9	Buzzer	Diameter × height, sound hole yes/no	
—	Decisions	Shape / layout / dial position / lid fixing / material	

With these, the base and lid STLs get regenerated to fit, ready to slice in Bambu Studio.